AuraZ

Sepsis early warning you can explain, trust, and run beside the data on IBM Z.

IBM Z Datathon 2026 · Technology for Good · github.com/saksham-2x7/IBM_Z

Research prototype on retrospective data, not a medical device.

The problem

- Sepsis is linked to about **11 million deaths a year** (WHO); every hour of delayed antibiotics lowers survival.
- ICU alarms such as SIRS fire so often that staff **ignore them**.
- Cloud AI moves **protected patient data** off-site and adds network delay.
- Hospitals need warnings that are **earlier, rarer, explained, and honest about uncertainty**, running beside the data.

The solution

- Scores every ICU patient **hourly**, predicts sepsis within 6 hours
- Plain-language **explanation** for every alert
- Conformal label: **sepsis not ruled out / uncertain / ruled out**
- Alert threshold under a **false-alarm budget**
- **AuraZ-R**: 300 trees, 108 features, runs on CPU
- **AuraZ-N**: neural network, runs on the Telum II accelerator
- **200-bed ward** view, **FHIR** endpoint, alert log
- 3,000-stay simulated demo + 14,635 real stays

Real ICU data results (PhysioNet 2019)

0.82

AUROC, held-out stays (95% CI 0.79–0.84)

0.79

AUROC without ICU-time features

95% vs 89%

septic patients caught, at SIRS's alarm rate

0.74

AUROC at a different hospital

On all 39,317 stays: AUROC 0.83, 96% vs 87% caught, other hospital 0.77–0.79. Our first model scored only **0.61** on real data, so we rebuilt it. We also report the drop at a new hospital: **each hospital must recalibrate**. Subgroup audit by sex and age is in the model card.

Running on a real IBM z17 (Telum II)

33×
AuraZ-N on the NNPA accelerator vs a plain CPU loop

1.5×
vs tuned OpenBLAS (1 thread); 1.2× vs both threads

0.8086
AUROC on the chip vs 0.8086 FP32

11,870
scores/s peak, 2 vCPUs, 0.27 ms median

32,768 patient-hours: **10.9 ms** on the accelerator vs 355 ms plain loop, 15.9 ms OpenBLAS. Honest limits: our tiny logistic model is **slower** on the accelerator, and the tree model stays on the CPU.

Architecture

Layer	What	Where
Data	PhysioNet 2019 real ICU stays; simulated look-alike cohort	ml/
Models	AuraZ-R trees, AuraZ-N neural network, conformal labels	Python, scikit-learn
Serving	Dependency-free service: /score_real, /fhir/score, alert log	IBM z17 (s390x)
Accelerator	AuraZ-N through IBM zDNN on the NNPA	C, Telum II
Clinician app	Live replay, 200-bed ward, validation, IBM Z panel	React + TypeScript
Quality	23 app tests, 6 server tests, 27 end-to-end checks, CI	Vitest, unittest, Actions

Honest evaluation

- Real-data first model: **0.61**; rebuilt model: **0.82**; at another hospital: **0.74**
- Uncertainty guarantee holds inside a hospital, **weakens across hospitals**
- Subgroup alert rates differ on small groups: flagged for follow-up
- Uses 37% of PhysioNet 2019, retrospective only, no race/ethnicity data
- Accelerator speed-up: 33× vs a plain loop, **1.2–1.5×** vs tuned OpenBLAS

Next

All 40k stays and validation at another hospital · per-hospital recalibration · close the 0.02 AUROC gap between the neural and tree models · HL7/FHIR subscriptions · silent-mode clinical pilot

AuraZ: explainable, honest about uncertainty, built to run beside the data on IBM Z.

Code: github.com/saksham-2x7/IBM_Z · Whitepaper, model card and accelerator report in /docs